[Part 2] Candidate Mechanism LEVEL Local Projections: Splitwise AR(1)-Innovation LPs without VIX as a Control

1. Data Overview

- **Database file**: primary `Q_Levels_Database.mat` (a single `table` variable, 160×36); `Q_Levels_Database.csv` kept as a CSV mirror fallback. The two files have been verified identical: same 36 columns, same 160 rows.
- **Frequency**: quarterly.
- **Database coverage**: 1986Q1 – 2025Q4 ($T = 160$ quarters). Unlike Part 3B, no early predictor binds the start: every series used here is available from 1986Q1 onward.
- **Sample mask**: the full database sample is used; no sub-period or outlier exclusion is applied.
- **Effective samples** (after $p_{\text{innov}} = 1$ lag for shock extraction and $p_{\text{lp}} = 4$ lags for the LP):
  - AR(1) innovation $z_t^{(s)}$: **1986Q2 – 2025Q4**, $n_{\text{AR}} = 159$.
  - LP at horizon $h = 0$: start at $t = p_{\text{lp}} + 1 = 5$ (**1987Q1**), end at $t = T - h = 160$ (2025Q4); roughly $n_{\text{obs}} \approx 156$ at $h = 0$, shrinking by one quarter per horizon to $n_{\text{obs}} \approx 144$ at $h = 12$ (subject to outcome-specific NaN trimming).

Sample-loss accounting

From 160 quarters of data, the LP at horizon h ends up using roughly $156 - h$ observations, less any extra trimming from missing outcome values. The drop happens in three stages:

#	Cause	Effect	Detail
1	AR(1) shock extraction (Stage 1)	$160 \to 159$	First usable shock at $t = 2$ (1986Q2). All 3 GPR series complete from 1986Q1, so $n_{\mathrm{AR}} = 159$ for each shock.
2	LP own-lags (Stage 2)	$159 \to 156$ at $h = 0$	Earliest feasible $t = p_{\mathrm{lp}}+1 = 5$ (1987Q1). LP loses 4 quarters total at the start; 1 already counted in step 1.
3	Future response y_{t+h}	$156 \to 156 - h$	Need $t + h \leq 160$. $h = 12 \Rightarrow n_{\mathrm{obs}} \approx 144$; endpoint slides from 2025Q4 down to 2022Q4.

Table 1. Variables used in Part 2 (No-VIX-Control specification).

Block	Code	Series / definition	Form	Source (typical)
GPR shocks (Stage 1)	`LGPR`	World GPR	log level	Caldara & Iacoviello (CI)
	`LGPRT`	Threat GPR	log level	CI
	`LGPRA`	Act GPR	log level	CI
Oil prices	`Real_WTI`	Real WTI crude oil price	level	EIA / FRED, deflated
	`Real_Gasoline`	Real gasoline price	level	EIA / BLS, deflated
Inflation expectations	`MICH`	Michigan 1Y expected inflation	%	U. Mich. Survey
	`SPF_inf`	SPF expected inflation (current)	%	Phila. Fed SPF
	`Two_PH_SPF_Inf`	SPF expected inflation (2-step ahead)	%	Phila. Fed SPF
Real activity	`r_c`	Real consumption (level)	log level × scale	BEA / FRED
	`r_i`	Real investment (level)	log level × scale	BEA / FRED
	`r_y`	Real GDP (level)	log level × scale	BEA / FRED
Risk	`lnVIX` (= log`VOX`)	log VIX/VOX	log level	CBOE / FRED
Macro controls	`Unemp`	Unemployment rate	%	BLS
	`FFR`	Effective Fed Funds rate	%	FRED

2. Model Specification

The script proceeds in two stages: **(i) extract a univariate AR(1) GPR innovation per shock series; (ii) estimate the dynamic level-change response of each outcome via a splitwise local projection.**

2.1. Stage 1 — Univariate AR(1) GPR innovations

For each shock series $s \in \{\text{World, Threat, Act}\}$, with $g_t^{(s)} \equiv \text{LGPR}_t^{(s)}$:

$$g_t^{(s)} \;=\; \mu^{(s)} \;+\; \sum_{j=1}^{p_{\text{innov}}} \phi_j^{(s)} g_{t-j}^{(s)} \;+\; u_t^{(s)}, \qquad p_{\text{innov}} = 1. \tag{1}$$

- Estimated by OLS on the in-sample window where every regressor is finite. All three series are complete from 1986Q1, so $n_{\text{AR}} = 159$ for each shock.
- **Shock**: residual standardised in-sample, $z_t^{(s)} \equiv (u_t^{(s)} - \bar{u}^{(s)})/\hat{\sigma}_{u^{(s)}}$, so a unit shock corresponds to one in-sample standard deviation of the AR residual.
- Rationale: **removes the one-quarter own-lag predictability of the GPR index so that $z_t^{(s)}$ is the news component of geopolitical risk relative to its own immediate past.**
- **Limitations relative to Part 3B's augmented shock. ★**
  This is the simpler "original" shock (univariate AR(1)) — it is *not* orthogonalised against oil-market or macro lags. A non-trivial portion of $z_t^{(s)}$ will therefore remain predictable from past oil prices, production, inventories, and inflation. In this dataset Eq. (1) yields $R^2 = 0.483$ / 0.470 / 0.516 for World / Threat / Act, respectively. For comparison, the augmented (ADL) variant in Part 3B raises the in-sample R^2 from ≈ 0.4–0.7 to ≈ 0.7–0.9. The Part 2 design is the appropriate baseline for showing what a textbook AR(1) shock implies about each candidate mechanism, before any augmentation.

Table 2. Lag-length choices and their justification.

Choice	Conventional rationale	Data-driven justification
$p_{\text{innov}} = 1$	GPR is a text-based index — noisy, with limited self-persistence beyond one quarter. Caldara–Iacoviello (2022, AER) likewise use AR(1) as the baseline.	× — no formal lag selection (AIC/BIC) in this script.
$p_{\text{lp}} = 4$	Quarterly data ⇒ 4 lags = 1 year of past information. Default in most quarterly macro forecasting work (Stock–Watson, Mertens–Ravn, etc.).	× — same as above.

2.2. Stage 2 — Splitwise local projections in level-change form

For each outcome $y \in \mathcal{Y}$ **(9 variables)** and **each shock** s **(3 series)**, and each horizon $h \in \{0, 1, \ldots, 12\}$ quarters, a separate single-shock LP is run:

$$y_{t+h} - y_{t-1} = \alpha_h^{(y,s)} + \beta_h^{(y,s)} z_t^{(s)} + \sum_{j=1}^{p_{\text{lp}}} \rho_{h,j}^{(y,s)} y_{t-j} + \sum_{j=1}^{p_{\text{lp}}} \gamma_{h,j}^{(y,s)} \text{LGPR}_{t-j}^{(s)} + \sum_{j=1}^{p_{\text{lp}}} \boldsymbol{\theta}_{h,j}^{(y,s)\prime} \mathbf{C}_{t-j} + \varepsilon_{h,t}^{(y,s)}. \tag{2}$$

(i) **LHS — level change, not level.**

- The dependent variable is the *change* $y_{t+h} - y_{t-1}$, measured from the period immediately before the shock to h quarters after.
- Therefore β_h **is by construction already a *level-change* response**. **Do not cumulate** β_h **over** h; cumulating would give the area under the change-response curve, which is not the level IRF.

(ii) **RHS structure.**

- $p_{\text{lp}} = 4$ — own lags of y ($j = 1, \ldots, 4$); lagged GPR *levels* $\text{LGPR}_{t-j}^{(s)}$ ($j = 1, \ldots, 4$); lagged controls $\mathbf{C}_{t-j}$ ($j = 1, \ldots, 4$).
- Control vector: $\mathbf{C}_t = (\texttt{Unemp}_t,\ \texttt{FFR}_t)$. **No VIX** (the no-VIX-control design choice).
- **No lagged** z**-values.** Past *shocks* are *not* added to the RHS — instead, past *GPR levels* carry the dynamic information. (Compare Part 3B, where 4 lags of $z_t^{(s)}$ appear on the RHS.) The two parameterisations are different but both control for "geopolitical-risk history": here through the level path, in Part 3B through the orthogonalised-shock path.
- Number of regressors: $k = 1 + 1 + p_{\text{lp}} + p_{\text{lp}} + p_{\text{lp}} \cdot |\mathbf{C}| = 1 + 1 + 4 + 4 + 4 \cdot 2 = 18$.

(iii) **Splitwise estimation.**

- Each shock enters its *own* LP.
  For one outcome y there are 3 separate regressions: one with z_t^{World}, one with z_t^{Threat}, one with z_t^{Act}. The lagged-level controls in each LP use the matching shock's own GPR level (LGPR, LGPRT, LGPRA respectively).
- Total regressions per figure cell:
  9 outcomes $\times$ 3 shocks $\times$ 13 horizons = 351. Note the multiplicative blow-up relative to a joint-LP design — *there is no Bonferroni correction applied.*

(iv) **Coefficient of interest.**

- $\beta_h^{(y,s)}$**: partial response of** $y_{t+h} - y_{t-1}$ **to a one-unit movement in** $z_t^{(s)}$**, holding lagged** y**, lagged GPR levels, and lagged controls fixed.**
- **All IRF figures here plot** $\widehat{\beta}_h \times 1$**, i.e. the response to a <u>1-σ GPR shock</u>**. This is the Caldara–Iacoviello (2022) convention. Magnitudes in this document are therefore *half* of those in Part 3B (which uses a 2-σ scale); double the numbers reported here when comparing with Part 3B figures.

(v) **Inference.**

- Estimation by **OLS; HAC long-run variance via a Newey–West Bartlett-kernel sandwich**.
  With finite-sample correction the variance is $\widehat{V} = \frac{n}{n-k}(X'X/n)^{-1}\,\widehat{\Omega}\,(X'X/n)^{-1}/n$, where $\widehat{\Omega}$ is the Bartlett-kernel HAC long-run variance of the score $x_t u_t$.
- **Bandwidth: Lazarus, Lewis, Stock & Watson (2018, JBES) Bartlett rule**, bw $= \lceil 1.3\sqrt{n} \rceil$ (capped at $n-1$). At Part 2's typical sample size of $n \approx 156$ at $h = 0$ this gives bw ≈ 17, *constant across horizons*. The LLSW rule is larger than both Stock–Watson (2010, $\lceil 0.75\, n^{1/3} \rceil \approx 4$) and the Jordà (2005) $\min(h+1,\, n-1)$ default (≤ 13 here), trading some asymptotic power for substantially better size in HAR inference.
- 90 % CI uses $z^* = \Phi^{-1}(0.95) \approx 1.645$.

3. Results

- **Left Col:** Response to the World GPR shock (grey 90% NW HAC CI band)
- **Right Col:** Threat (blue) and Act (red)
- **Reference Line:** Dashed zero line for reference
- **X-axis:** Horizon h in quarters (0 to 12)
- **Y-axis:** Level change $y_{t+h} - y_{t-1}$ in the source units of y
- **Shock Size:** All responses are to a 1-σ GPR shock

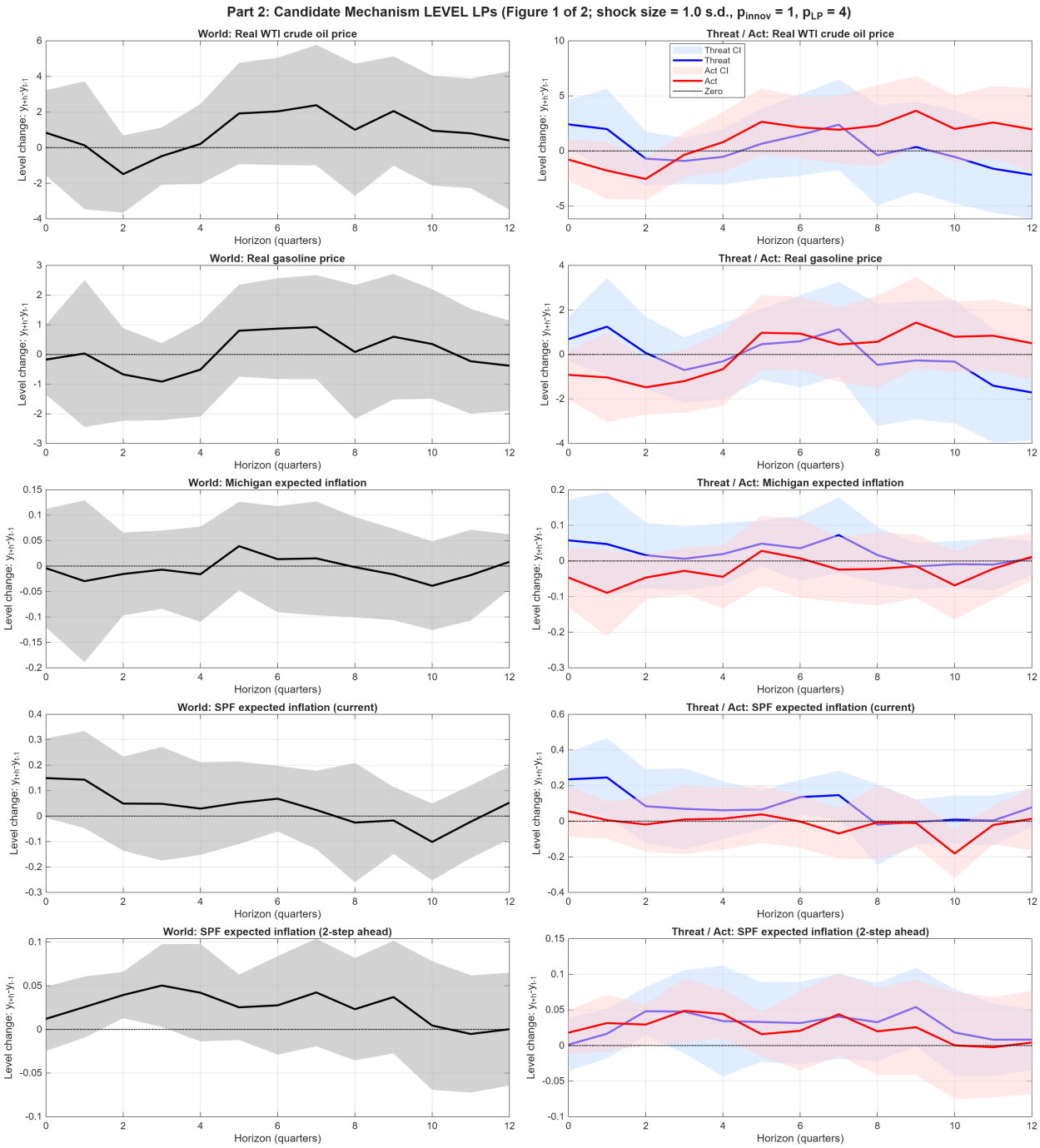

Figure 1. Part 2 mechanism level LPs — Figure 1

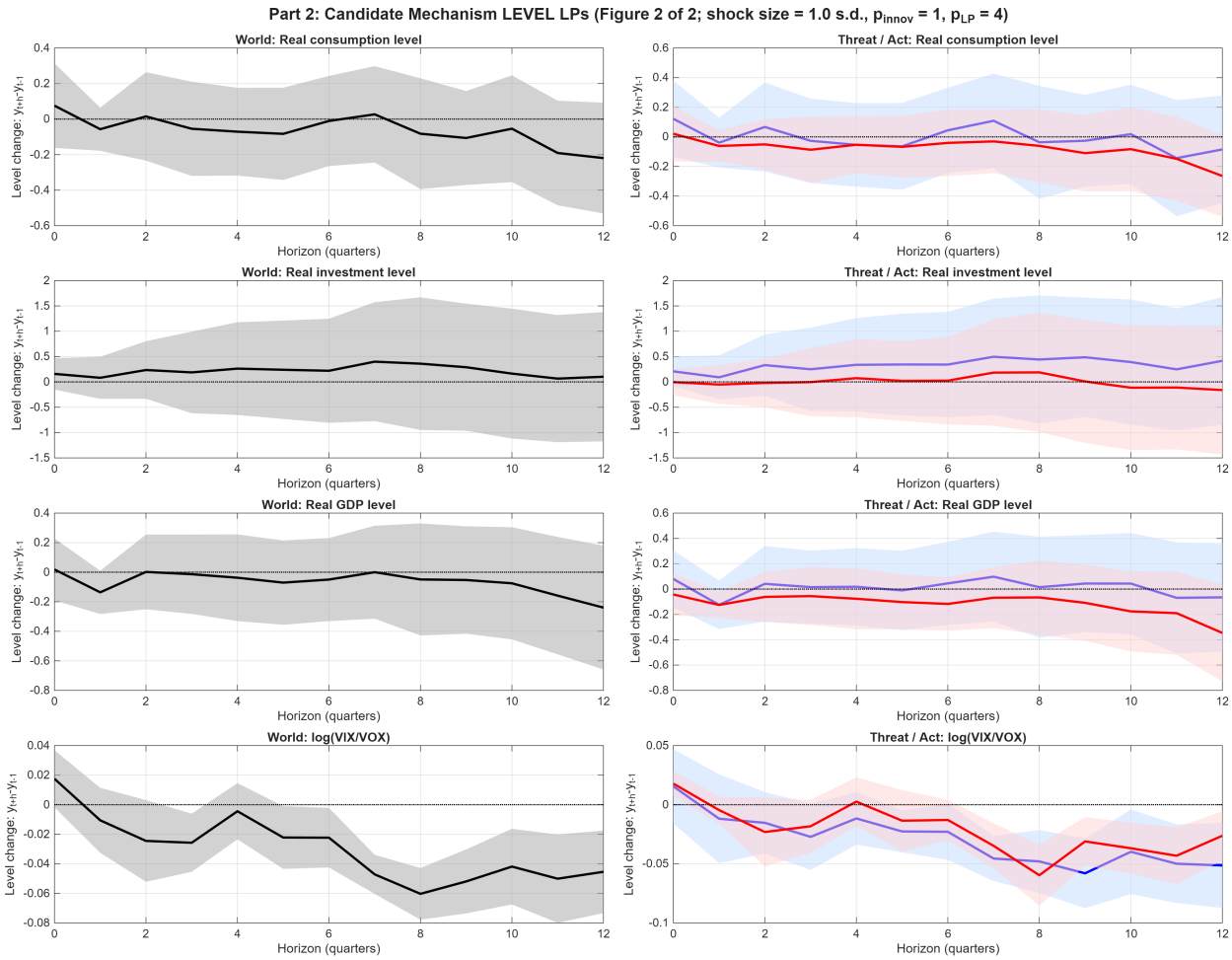

Figure 2. Part 2 mechanism level LPs — Figure 2

3.1. Reading the IRF panels

Table 3. Decoding the column-pair structure of each row.

Comparison	Pattern observed	Reading
Threat vs. Act (right tile)	Lines co-move	Mechanism reacts to risk *perception* alone; realisation is noise.
	Lines diverge / opposite signs	Realised events carry incremental information — markets / expectations price the two distinctly.
World vs. split sub-indices (cross-row)	World aligned with Threat	Aggregate response driven by anticipation news.
	World aligned with Act	Aggregate response driven by realised disruption.
	World decoupled from both	Aggregate captures composition / weighting effect that sub-indices miss.

Table 4. LP IRF reading tips (memo-style reference).

Tip	What it means
$\widehat{\beta}_h$ is already cumulative	LHS is $y_{t+h} - y_{t-1}$, so the curve traces the *level deviation from baseline*. **Do NOT sum $\widehat{\beta}_h$ across h.**
Units = source units of y	pp for `MICH`, `SPF_inf`; log points for `lnVIX`; real-units for `r_c` / `r_i` / `r_y`; dollars for `Real_WTI` / `Real_Gasoline`.
Decay $\to 0$	Effect is *transitory*: level returns to baseline.
Flat at non-zero	*Permanent* level shift.
Grows with h	*Accumulating* pass-through (rare; possible).
Oscillates around zero	Noise-dominated; treat as null.
Band excludes 0 at horizon h	Significant at 10% by *marginal* NW test *at that horizon only* — not a joint test.
Multiple-testing risk	351 marginal tests $\Rightarrow$ expect $\sim$35 false rejections at nominal 10%. **Trust patterns across horizons + related outcomes, not isolated points.**
Bands widen with h	Mechanical: n_{obs} shrinks by 1 per horizon. Don't over-weight tight $h = 0$ band.
Use "response" not "effect"	AR(1) shock partially predictable from past oil/macro; identification is weaker than Part 3B's augmented version.

3.2. Channel-by-channel observations

	World		**Threat**		**Act**	
Outcome	$h = 0$	$h = 4$	$h = 0$	$h = 4$	$h = 0$	$h = 4$
`Real_WTI`	+0.84	+0.21	**+2.41**	−0.53	−0.78	+0.80
`Real_Gasoline`	−0.17	−0.51	+0.69	−0.31	−0.92	−0.66

Oil-price channel (Figure 1, rows 1–2).

- **Threat-WTI $h = 0$ peak (+2.4) is the largest single magnitude in the entire study** — the only finding that visually excludes zero by the 90% band.
- **Sign reversal Threat (+) vs. Act (−) on WTI at $h = 0$.** Markets price WTI up on anticipation, down on realisation — realised events appear already-priced or signal demand destruction.
- **Gasoline lags WTI and shows a weaker, mixed pattern.** Slower refining-margin pass-through dominated by other factors in this sample.

	World		Threat		Act	
Outcome (pp)	$h=0$	$h=4$	$h=0$	$h=4$	$h=0$	$h=4$
`MICH`	−0.004	−0.016	+0.058	+0.019	−0.046	−0.045
`SPF_inf`	+0.149	+0.030	**+0.235**	+0.061	+0.056	+0.014
`Two_PH_SPF_Inf`	+0.012	+0.042	+0.001	+0.034	+0.018	+0.044

Inflation-expectations channel (Figure 1, rows 3–5).

- **Threat moves SPF current (+0.24 pp) more than household Michigan (+0.06 pp).** Professionals revise current-year forecasts up on Threat news; households react less.
- **Two-step-ahead SPF essentially does not move** ($|\widehat{\beta}_h| \approx 0$ across all horizons and all shocks). **Cleanest evidence of medium-term expectation anchoring** in the document — Fed credibility intact.
- **Act responses uniformly small.** Realised events do not generate fresh upward pressure on expectations beyond what was already priced.

	World		Threat		Act	
Outcome	$h=0$	$h=4$	$h=0$	$h=4$	$h=0$	$h=4$
`r_c` (cons.)	+0.076	−0.071	+0.122	−0.054	+0.021	−0.053
`r_i` (inv.)	**+0.158**	**+0.262**	**+0.208**	**+0.337**	−0.005	+0.073
`r_y` (GDP)	+0.019	−0.038	+0.080	+0.018	−0.043	−0.076

Real-activity channel (Figure 2, rows 1–3).

- **Investment anomaly.** $\widehat{\beta}^{\mathrm{r_i}}$ is positive across all shocks and grows with h. Contradicts standard real-options prediction (GPR $\uparrow \Rightarrow$ wait-and-see $\Rightarrow$ investment $\downarrow$).
- **Three candidate explanations:** (i) energy-capex booms in oil-supply-shock-heavy sample dominate uncertainty effect; (ii) lagged-GPR-level RHS absorbs persistent risk differently than lagged-shock alternative; (iii) quarterly aggregation smooths within-quarter pauses.
- **Consumption muted, GDP near zero.** No robust consumption channel; GDP responses appear to net out across components.

	World		Threat		Act	
Outcome (log pts)	$h = 0$	$h = 4$	$h = 0$	$h = 4$	$h = 0$	$h = 4$
`lnVIX`	+0.018	−0.004	+0.016	−0.012	+0.018	+0.003

Financial-risk channel (Figure 2, row 4).

- **lnVIX response is weak** ($\sim +1.8\%$ in VIX at $h = 0$, decaying to ~ 0). Comparable across all three shocks.
- **If VIX is the principal financial-market mediator, this channel does not dominate.** Either GPR captures a different dimension than option-implied volatility, or VIX responds within-quarter and gets averaged out at quarterly frequency.
- **Read jointly:** muted lnVIX + strong Threat WTI / SPF ⇒ **GPR transmits through commodity and expectations channels more than equity-risk pricing** in this sample.

4. Implications

(i) **Methodological — what the design lets us claim.**

- **Level-change LP form gives a one-shot level IRF.**
  The form $y_{t+h} - y_{t-1} = \alpha_h + \beta_h z_t +$ controls produces an IRF whose units match the LHS source units . **No manual cumulation needed.**
- **Splitwise design avoids multicollinearity but loses joint testing.**
  World / Threat / Act are highly correlated by construction (the latter two are nested components of the first). Putting all three on the same RHS would inflate standard errors. **Splitwise LPs cleanly identify each shock's marginal response, but cannot answer "after controlling for Threat, does Act add anything?"** — that requires a companion joint specification with all three on the RHS, which is left for future work.
- **Lagged GPR levels (not lagged shocks) on RHS.**
  A Wold-style level-VAR conditioning: the LP partials out the entire level path of geopolitical risk, not just recent innovations. Appropriate when the shock is constructed from the same series as the lagged controls. Ensures the contemporaneous coefficient on z_t is not contaminated by the persistent component of GPR.
- **Identification is weak under AR(1).**
  The identifying assumption is $z_t^{(s)} \perp \{y_{t-j}, \mathrm{LGPR}_{t-j}, \mathbf{C}_{t-j}\}_{j\geq 1}$. Under AR(1) shock construction this is *weaker* than under Part 3B's augmented (ADL) construction — $z_t^{(s)}$ may still be predictable from past oil prices, inflation, or activity (i.e., $R^2 \approx 0.48$, so 48% of GPR variation is past-explained, leaving the residual potentially still

containing past-correlated mechanical components). Hence "*response*" rather than "*effect*".

- **HAC inference choice.**
  Newey–West with LLSW (2018) Bartlett rule, bw $= \lceil 1.3\sqrt{n} \rceil \approx 17$ at typical $n \approx 156$, constant across horizons. This is the published recommendation for HAR inference, size-improving relative to Stock–Watson (2010, bw ≈ 4) and Jordà (2005, bw ≤ 13). Bands here are wider than smaller bandwidths would give — power cost is acceptable since Part 2's role is descriptive, not significance-driven.

- **Sample size shrinks predictably with horizon.**
  At $h = 0$, $n_{\text{obs}} \approx 156$; at $h = 12$, $n_{\text{obs}} \approx 144$. Differences across outcomes come only from outcome-specific NaNs (none in this sample for the 9 outcomes used). So band widening at long h is mostly mechanical.

(ii) **Findings — pattern-by-pattern.**

Pattern	**Evidence**	**Implication**
Threat dominates contemporaneous responses	$\lvert\widehat{\beta}_0^{\text{Threat}}\rvert > \lvert\widehat{\beta}_0^{\text{Act}}\rvert$ for almost every outcome.	Markets / expectations price GPR on *anticipation*, not materialisation. Consistent with forward-looking asset pricing.
Investment anomaly	$\widehat{\beta}_h^{\text{r_i}} > 0$ across all shocks, growing with h ($+0.16 \to +0.26$ World; $+0.21 \to +0.34$ Threat).	Contradicts real-options theory. Most important target for Part 3B robustness check.
Inflation expectations anchored beyond 1Q	Two-step-ahead SPF $\lvert\widehat{\beta}_h\rvert \approx 0$; only current-quarter responds, only under Threat.	GPR transmits to prices via cost-channel (oil), not via expectation destabilisation. Fed credibility intact in this sample.
VIX channel is weak	$\widehat{\beta}_0^{\text{lnVIX}} \approx 0.018$ (small) for all shocks; decays to ~ 0 by $h = 4$.	GPR transmits through commodity / expectation channels more than equity-risk pricing. The "no-VIX-control" design choice has modest practical consequences here.
World $\neq$ blend of Threat + Act	WTI, gasoline: World aligns with one sub-index, not a smooth average.	Aggregate index has compositional heterogeneity; not just a weighted sum.
Statistical caveat	Only Threat-WTI $h = 0$ peak ($+2.4$) visually excludes zero across multiple horizons. Most patterns have CI bands crossing zero.	Treat patterns as descriptive characterisations, not established effects, pending Part 3B's augmented-shock check + a joint-LP companion test.

(iii) **Substantive — what the design lets us say.**

Channel	**Variables**	**What we test**
Oil-price	`Real_WTI`, `Real_Gasoline`	Whether GPR shocks move real oil-product prices over horizons of up to 3 years.
Inflation expectations	`MICH`, `SPF_inf`, `Two_PH_SPF_Inf`	Whether GPR shocks shift household and professional inflation expectations — the *psychological* pass-through into expected prices, distinct from the realised oil-price response.
Real activity	`r_c`, `r_i`, `r_y`	Whether GPR shocks contract consumption, investment, and output. Investment is the channel most likely to react under real-options logic (uncertainty raises the option value of waiting).
Financial-risk	`lnVIX`	Whether GPR shocks raise option-implied equity volatility. This is precisely the channel *absorbed* when `lnVIX` is used as a control; including it only as an outcome is what makes the no-VIX-control design informative.
Threat vs. Act decoupling	—	Whether the mechanism reacts to *anticipation* (Threat) or *materialisation* (Act). Investment / inflation expectations are leading Threat-dominant candidates; production / realised inflation are leading Act-dominant candidates.

(iv) **Caveats and next steps.**

Caveat	**Implication**	**Suggested next step**
Shock size is 1σ	Half the convention in Part 3B (2σ).	Harmonise scale (re-plot Part 3B at 1σ or Part 2 at 2σ) or annotate clearly in every caption.
AR(1) shock is partially predictable	$z_t^{(s)}$ still predictable from lagged oil/macro variables; conflates "news" with mechanical projection.	Re-run Part 2 with Part 3B's augmented shock as a robustness check.
No VIX control by construction	Common driver of GPR and VIX is *not* partialled out; responses are *net of* GPR-VIX correlation.	Intentional — flag in interpretation rather than fix.
Splitwise design has no joint test	Cannot tell which of (World, Threat, Act) carries incremental signal once the others are controlled for.	Joint LP with all three shocks on the RHS, or a Threat-vs-Act contrast with a single dummy.
Sample includes structural events	Oil-price collapses 1986 / 2014 / 2020, GFC, COVID, 2022 Russia–Ukraine war may drive estimates.	Sub-sample stability (pre-2020, pre-COVID) and state-dependent IRFs (recession vs. expansion, high vs. low VIX).
HAC bandwidth follows LLSW (2018)	Matches Part 3B's bandwidth choice in spirit but not in formula (Part 3B uses $\lceil 0.18n \rceil$).	Side-by-side overlay against the Jordà (2005) $\min(h+1, n-1)$ bandwidth.
No formal lag selection	$p_{\text{innov}} = 1$, $p_{\text{lp}} = 4$ are imposed.	AIC/BIC-driven selection or sensitivity to $p_{\text{lp}} \in \{2, 4, 8\}$.
Five-row tiled layout is dense	Hard to compare mechanism blocks at this density.	Split into per-channel figures (oil prices / inflation expectations / real activity / VIX) with consistent y-axis scaling.